

João Pedro Carvalho dos Santos

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Summary

Full Stack Developer with 2.5 years of experience and strong command of TypeScript. I have expertise in modern Front-end (Next.js, Remix, Astro) and comprehensive experience in Back-end (Python, Node.js), Mobile (React Native, Expo), and DevOps for cloud solution deployment. I am the creator and maintainer of StarDust, an Open Source platform for teaching programming logic. Currently pursuing a degree in Cross-platform Software Development at Fatec São José dos Campos, I serve as CTO of the Ctrl Alt Del group, leading a collective of developers in building technological solutions for academic and corporate challenges.

Professional Experience

Design Líquido | Internship

— Full Stack Web Developer | May/2025 – Dec/2026

Worked as the lead developer of the StarDust educational web platform, which teaches programming logic to IT beginners through gamification.

- ✓ Developed end-to-end (Full Stack) features, from interface prototyping to database and API integration.
- ✓ Responsible for the system architecture, ensuring component decoupling for high scalability and easy maintenance.
- ✓ Implemented CI/CD pipelines and automated test coverage (unit and integration) to ensure continuous and stable deliveries.
- ✓ The platform was adopted as a teaching support tool for incoming classes at Fatec São José dos Campos.
- ✓ Capstone project for the Systems Development technical program at Etec, highly praised by the examination committee.
- ✓ Implemented monitoring and observability routines for error tracking and production availability assurance.

Skills: Design Patterns, Clean Architecture, Cloud Computing, User Experience (UX), Databases, SOLID, Clean Code, CI/CD

Tools: Next.js, TailwindCSS, TypeScript, Supabase (PostgreSQL), Inngest, Hono.js, Remix, Jest, Heroku, GitHub Actions, Sentry.io, Delégua

Access link: <https://stardust-app.com.br>

Ancra | Freelance

— Web Developer | Nov/2024 – Dec/2026

Responsible for the full development of a news portal aimed at promoting the dissemination of social inclusion projects and regional content.

- ✓ Integrated the application with Sanity.io for global CDN content distribution, creating an intuitive Rich Text editing experience that gave full autonomy to columnists and editors.
- ✓ Applied SEO strategies (meta tags, alt attributes, semantic HTML, and image optimization), maximizing indexing and organic reach of the blog.
- ✓ Automated the drafting of CMS posts using an AI agent distributed across multiple smaller sub-agents.

Skills: User Experience (UX), SEO Optimization, CDN

Tools: Astro.js, TailwindCSS, TypeScript, Sanity.io, FastAPI, Python, Agno, Google Cloud Platform

Access link: <https://pulodogatonews.com.br/>

Sertton Industrial | Freelance

— Mobile Developer | Sep/2023 – Dec/2023

Worked as lead developer throughout the complete lifecycle of a mobile sales application for large commercial vehicle body parts.

- ✓ Accelerated development by applying advanced context engineering techniques to systematically assist in AI-driven full application development.
- ✓ Used MVVM architecture pattern to ensure decoupling between business logic and interface, facilitating maintenance and testability.
- ✓ Responsible for interface prototyping, prioritizing usability and modern design principles to ensure an intuitive user experience.
- ✓ Integrated with the Yampi e-commerce platform REST API and developed a custom serverless function for shipping cost calculation.
- ✓ Automated the entire app publishing and update process on Google Play Store, eliminating manual processes and speeding up new version delivery (CD).

Skills: User Experience (UX), AI-Assisted Development, Software Architecture, API Integration, Clean Code

Tools: React Native, Expo, TypeScript, Jest, Cursor, Codex, Figma, Zustand, MMKV, GitHub Actions, Yampi API, Axios

Academic Projects

Smart Farming | Fatec - 1st Semester of 2024

Group development of a web application for telemetry and monitoring of a Smart Greenhouse on the university campus.

- ✓ Developed the integration layer with ESP microcontrollers, implementing automatic real-time sensor data ingestion to the database.
- ✓ Created dashboards with interactive charts for climate analysis and decision-making, using HTMX for dynamic interface updates without page reloads.
- ✓ Implemented automated routines (Python/Cron scripts) for daily data backup compression and cloud upload, ensuring information security.
- ✓ Complete containerization of the application and database using Docker, facilitating deployment and environment replication.

Skills: Agile Scrum Methodology, Requirements Analysis, Backlog Management, Teamwork

Tools: Python, Flask, HTMX, MySQL, Docker, TailwindCSS, HyperScript, IoT (ESP)